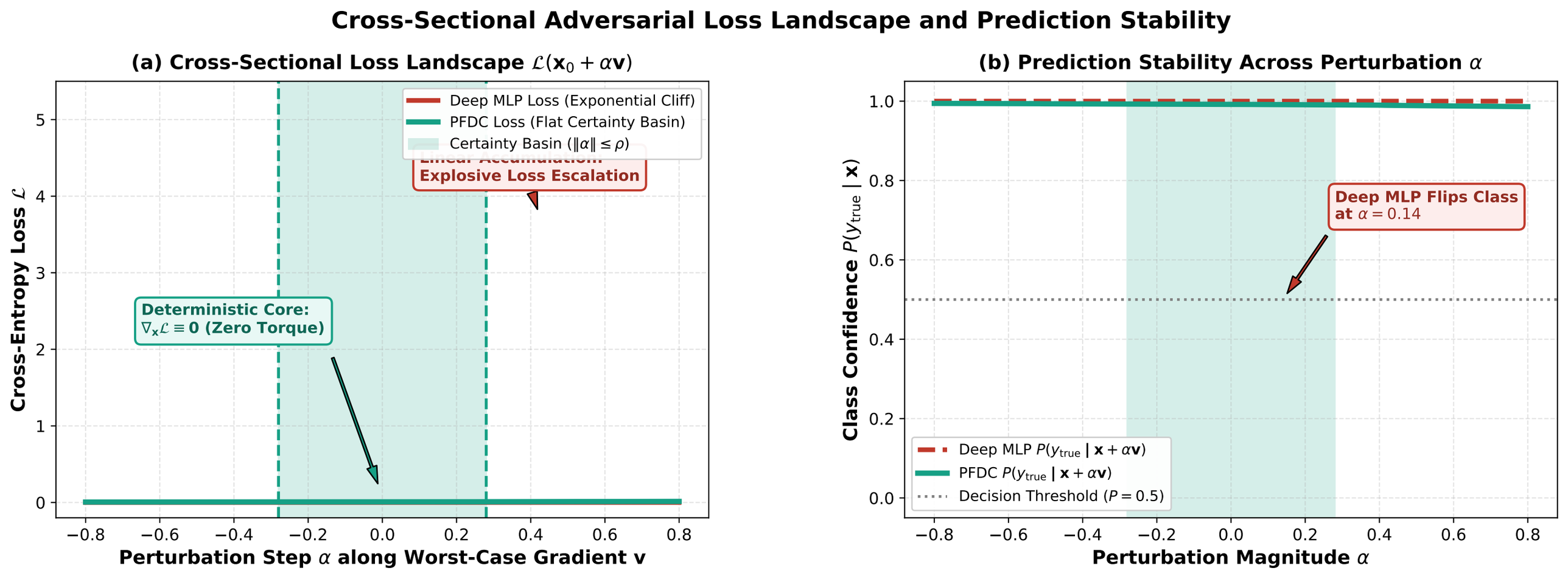
Cross-Sectional Adversarial Loss Landscape and Prediction Stability
(a) Cross-Sectional Loss Landscape $\mathcal{L}(\mathbf{x}_0 + \alpha\mathbf{v})$
Deep MLP Loss (Exponential Cliff)
PFDC Loss (Flat Certainty Basin)
Certainty Basin ($\|\alpha\| \leq \rho$)
Explosive Loss Escalation
Deterministic Core:
$\nabla_{\mathbf{x}}\mathcal{L} \equiv \mathbf{0}$ (Zero Torque)
Cross-Entropy Loss $\mathcal{L}$
Perturbation Step α along Worst-Case Gradient $\mathbf{v}$
(b) Prediction Stability Across Perturbation α
Deep MLP Flips Class at $\alpha = 0.14$
Deep MLP $P(y_{\text{true}} \mid \mathbf{x} + \alpha\mathbf{v})$
PFDC $P(y_{\text{true}} \mid \mathbf{x} + \alpha\mathbf{v})$
Decision Threshold ($P = 0.5$)
Class Confidence $P(y_{\text{true}} \mid \mathbf{x})$
Perturbation Magnitude α